Examiner only

8. Germany is building the world's first system in which wind turbines are combined with a hydroelectric pumped storage system.

(a) State the main disadvantage of relying on energy from wind turbines. [1]

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(b) When the output from the wind farm is not required by the German National Grid, the energy from the wind is used to pump water from a low level to a higher level.

The water at the higher level can then be used to generate electricity when required, just as in a conventional hydroelectric pumped storage system.

When generating, the power output from the hydroelectric pumped storage system is 16 MW.

The mean power used by homes in Germany was 0.43 kW in 2017 (which was the lowest on record up to that time).

(i) Suggest a possible reason why the 0.43 kW figure is lower than for any previous year. [1]

....

....

Examiner only

(ii) Calculate the number of homes in Germany in 2017, that could be supplied using the power from the hydroelectric pumped storage system. [3]

number of homes = ...

(c) The wind turbines have a maximum power output of 13.6 MW.

They provide energy to the water when slowly pumping it to a higher level.

When needed, the hydroelectric pumped storage system can quickly generate electricity.

Scientists state it is possible to get a greater **power** from the hydroelectric pumped storage system (16 MW) than the power supplied to it by the wind turbines.

By considering the equation:

$$\text{power} = \frac{\text{energy transferred}}{\text{time}}$$

explain why the claim is true. [2]

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END OF PAPER